



वित्तीय सेवाएं विभाग
DEPARTMENT OF
FINANCIAL SERVICES



सेन्ट्रल बैंक ऑफ़ इंडिया
Central Bank of India

Cyber Security PSBs Hackathon Series – 2026. An Initiative by DFS & IBA via PSBs, hosted at MNNIT Allahabad, Prayagraj.

Team : Mango

1. Title of the Proposed Idea:

AI-Driven Behavioral Analysis and Adaptive Fraud Detection System

2. Problem Statement :

- a) Current internet banking websites and environments lack modern up to date cyber security and detection systems
- b) Traditional authentication mechanisms such as OTPs, PINs, and biometrics alone may not be sufficient against sophisticated malware, session hijacking, credential theft, and behavioral spoofing attacks
- c) Many existing banking systems still lack sufficiently adaptive and context-aware behavioral analysis and accurate real-time transaction evaluation mechanisms

3. Proposed Solution :

- a) Behavioural tracking and location as well as IP monitoring before login to evaluate risk score
- b) Using dynamic authentication measures for users for more secure logins
- c) Using Advanced Classification and anomaly detection ML models along with synthetic fraud generation framework inspired by adversarial learning models for evolving cyber security scenario



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- d) Creating an end to end flag mechanism on transaction and behavioural tracking utilizing risk score evaluation to detect anomalies and possible hackers

4. PI Category :

Hybrid Software and Cloud Centric Implementation for ease of integration with current Net Banking systems

5. Technical Approach & Methodology:

This Idea we propose is currently in development through a modular approach -

- 1.prior to the login we monitor the users behavioural interaction patterns their IP ,location ,device signatures , VPN usage to judge if an actual user is present or not
2. After the login we closely monitor the transactions and transaction history to conclude if any transaction is valid or not
3. We are using a Training data set from **IEEE-CIS Fraud Detection** to train our ML model on multiple parameter listed as well as far more parameters for our procedurally generative adversarial Model that we test against our flagging model to check for its accuracy



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4. By using the data gathered we evaluate a risk score for every user and take preventive measures according to it such as triggering preventive security measures such as alert escalation, secondary verification, temporary transaction holds, and bank/user notifications.

6. Expected Outcome :

1. We believe by implementing modern adaptive and advanced systems to safe keep our accounts we can significantly reduce the probability of fraudulent transactions and incidences of cyber thefts
2. Our Solution can increase the accuracy and seamless detection of fraudulent activity by using proven preventive measures

7. Impact and Feasibility :

1. Our solution directly impacts banks as it is specifically catered to
The Net Banking audience
2. Our solution is an end to end engine which could be easily deployed in close cooperation with any net banking service being as modular and scalable as the bank wants it to be
3. Our model is technically feasible and easy to implement as we just have to run it as an extension to the existing infrastructure
4. Our only concern is available data to train our Model for India users